



Tree New Bee: Final Portfolio

Sammy Lee, Qingwan(Cecelia) Cheng, Joanne “Jo” Lee

Team Introduction



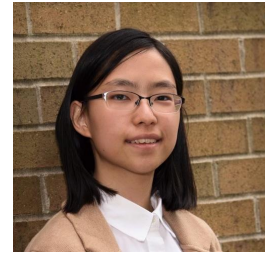
Sammy Lee

BC '23 (Senior)
CS BS, Studio Art
Minor
Malden, MA



Cecelia Cheng

BC '23 (Senior)
CS + Applied
Psych BA,
Cyber Minor
Nanjing, China

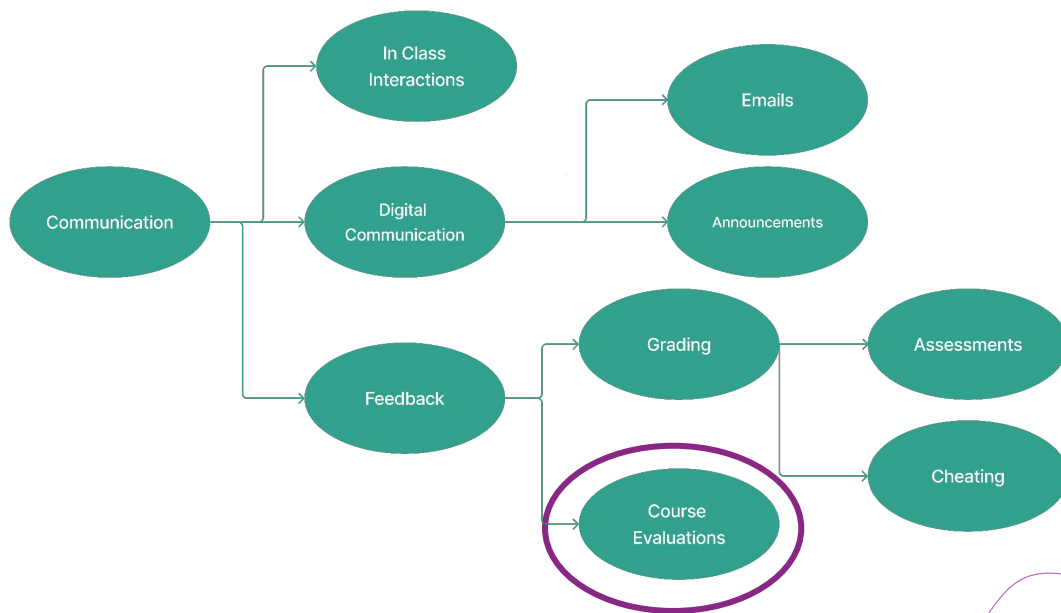


Jo Lee

BC '24 (Junior)
CS BS, Theology
Minor
Weston, MA

Problem Domain

Improving teaching experience for CS professors at Boston College



Solution Idea

HMW: How might we encourage honest, constructive feedback?

Solution Idea: Have customizable survey platform that utilizes a tagging system to help dissatisfied students form specific, constructive feedback.

Q4 Overall, how would you rate this course? ✎

Negative Comment	Positive Comment
Hard to follow ✕	Slides are easy to follow ✕
Boring ✕	Interesting topics ✕
Not practical in real world ✕	Practical knowledge ✕
Poor explanation ✕	Clear explanation ✕
Low engagement ✕	Active classroom Environment ✕
+	+

Question with Tags as Answers

Heuristic Evaluations Report Summary

- 3.2 Difficult navigation
- 2.9 No error messages for major changes
- 2.5 No documentation/ explanations
- 1.6 Confusing survey results page
- 1.8 No templates
- 1.7 No feedback after actions
- 1.6 Confusing meaning of colors
- 1 No grouping of elements on page

Q4 How would you describe the lectures?

Hard to follow Boring Lectures were fine Slides are easy to follow

Too slow Too fast Engaging

Other +

Q5 How would you describe the homework assignments?

Homework didn't match lecture materials Homework was fine Clear homework instructions

Confusing homework instructions Homework reinforced lecture materials

Tedious requirements for submissions Minimum Requirements

Difficult homework Easy homework

Other +

Q6 How would you describe the exams?

Confusing questions on exam Manageable Exam Matches with lecture materials

Difficult exam Clear questions on exam

Didn't match lecture material Easy exam

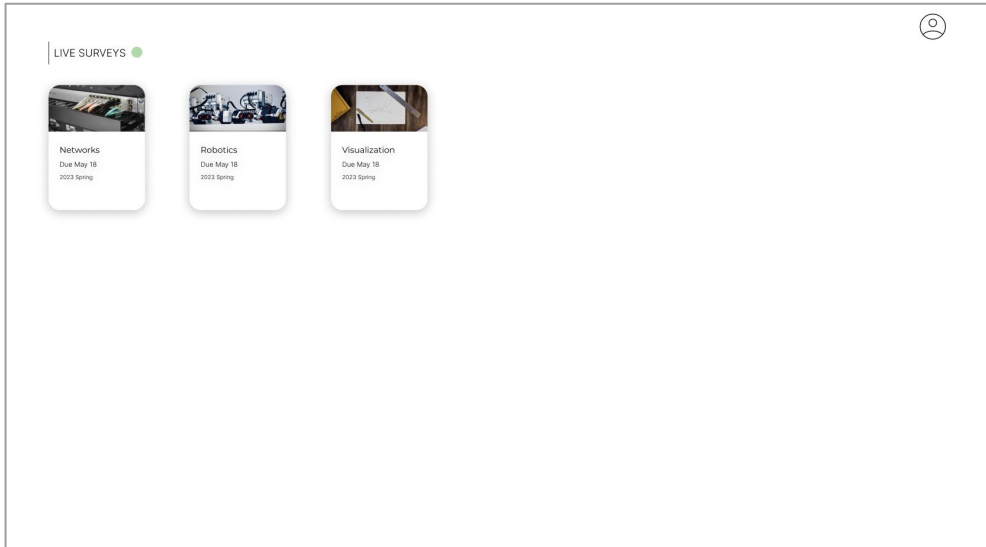
Other +

Back Submit

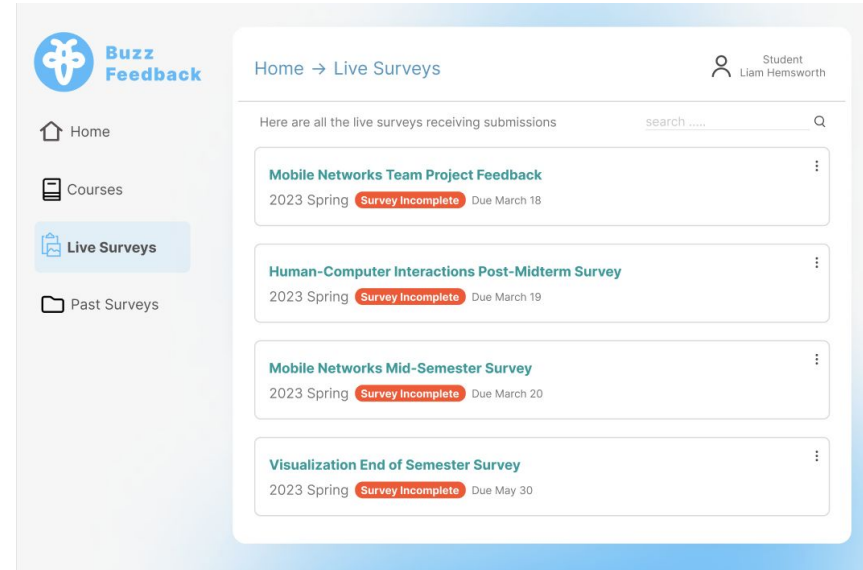
Difficult Navigation

Solution: Breadcrumbs + menu on the side

Before - Survey outline page



After - Survey outline page



Difficult Navigation

Solution: Breadcrumbs + menu on the side

Before - Question page

Q1 Overall, how would you rate this course?

☐ Excellent
☐ Very Good
☐ Good
☐ Fair
☐ Poor

Q2 Overall, how would you rate this instructor?

☐ Excellent
☐ Very Good
☐ Good
☐ Fair
☐ Poor

Q3 Overall, how useful were the resources (lecture recordings, textbook readings)?

☐ Extremely Useful
☐ Very Useful
☐ Somewhat useful
☐ Not so useful
☐ Not at all useful

[Next](#)

After - Question page

Buzz Feedback

Home → Live Surveys → Visualization End of ... Student: Liam Hemsworth

Visualization End of Semester Survey 2023 Spring Due 05/30/2023

Q1 Overall, how would you rate the course?

☐ Excellent
☐ Very Good
☐ Good
☐ Fair
☐ Poor

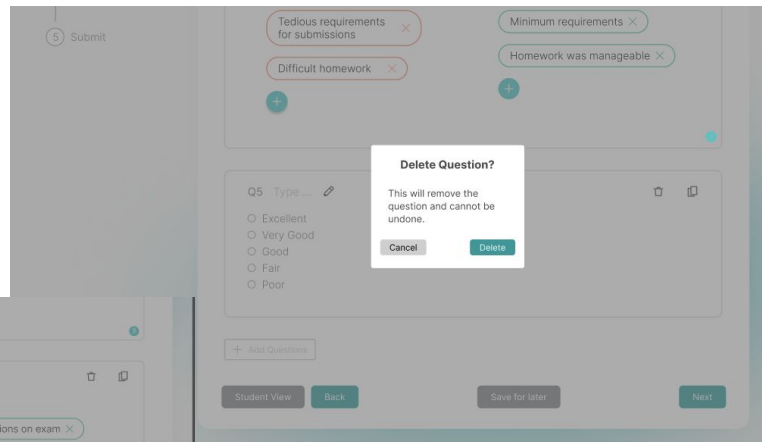
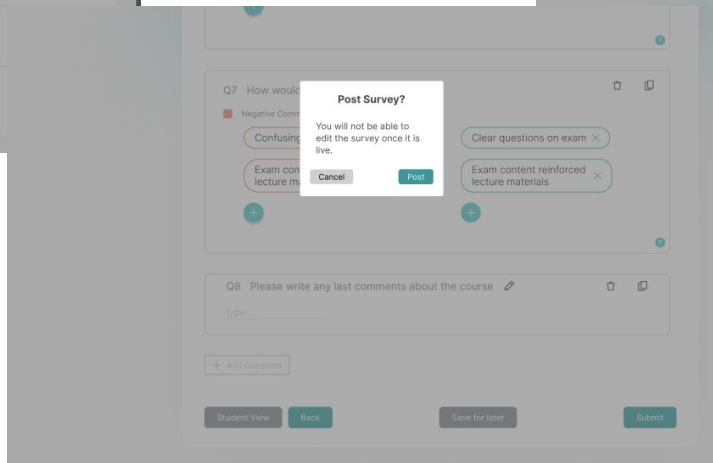
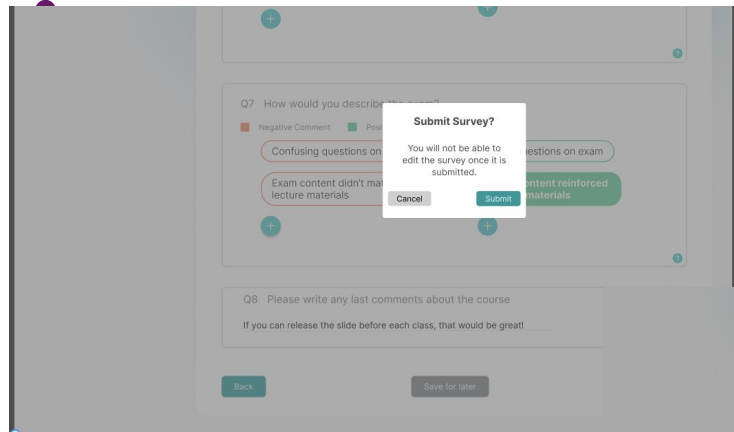
Q2 Overall, how would you describe the course?

☐ Negative Comment ☐ Positive Comment

[Back](#) [Save for later](#) [Next](#)

No Error Messages for Major Changes

Solution: Added error messages for deleting/posting/submitting



No Documentation / Explanations

Solution: Added little '?' buttons that bring popup explaining a feature

Q2 Overall, how would you describe the course? 

 Negative Comment  Positive Comment

Hard to follow 

Boring 

Not practical in real world 

Poor explanation 

Low engagement 

Slides are easy to follow 

Interesting topics 

Practical knowledge 

Clear explanation 

Active classroom Environment 

Q2 



Tagging System

Q4 Overall, how would you rate this course?

Hard to follow 

Boring 

Not practical in real world 

Poor explanation 

Low engagement 

Slides are easy to follow 

Interesting topics 

Practical knowledge 

Clear explanation 

Active classroom Environment 

Tags are key words or phrases that describe a theme to a topic. Users can select the tags they think describes their answers to a the question, and can also create new ones of their own.

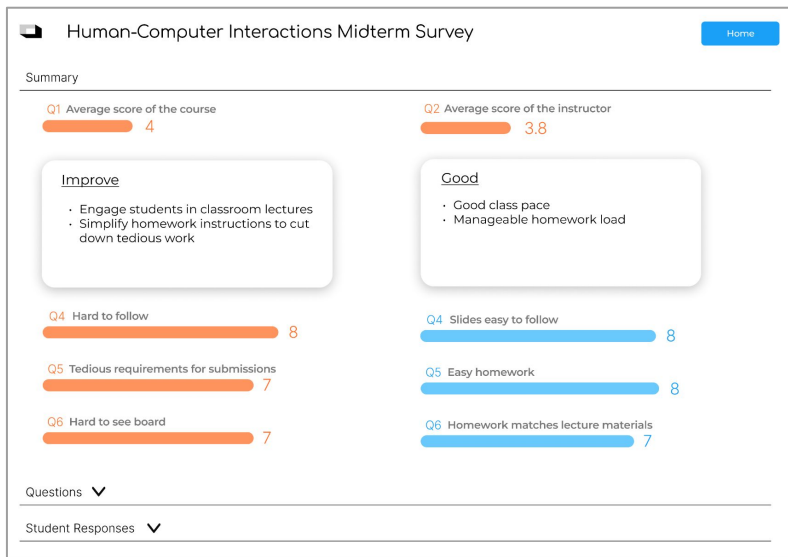




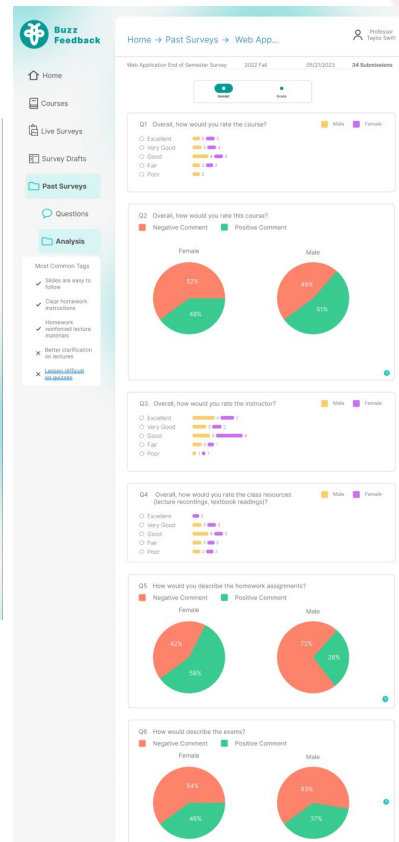
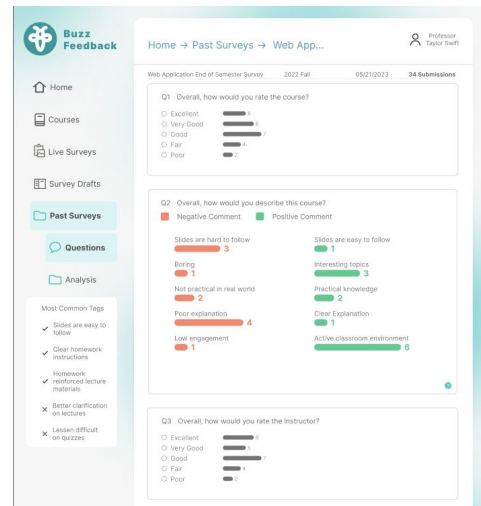
Confusing Survey Result Analysis

Solution: Added legend to explain colors' meanings

Before



After



No Templates

Solution: Added template option for creating survey

puter Interactions Post-Midterm Survey

Survey Incomplete Due March 19

4 / 28
Submissions

orks Mid-Semester Survey

Survey Incomplete Due March 20

19 / 25
Submissions

orks Team Project Feedback

Survey Incomplete Due March 30

1 / 25
Submissions

Create survey +

puter Interactions Post-Midterm Survey

Survey Incomplete Due March 19

4 / 28
Submissions

orks Mid-Semester Survey

Survey Incomplete Due March 20

19 / 25
Submissions

orks Team Project Feedback

Survey Incomplete Due March 30

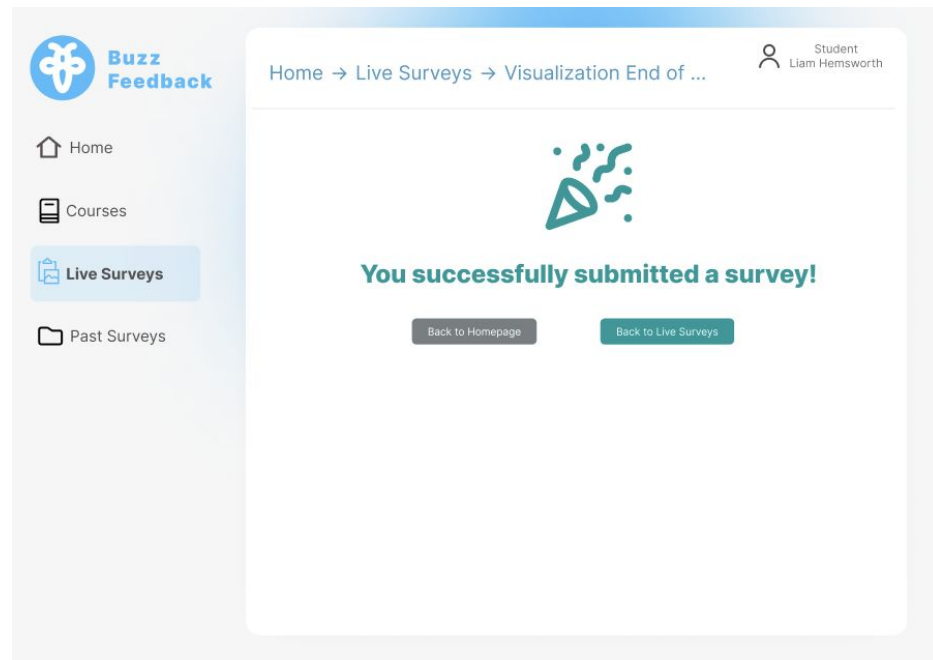
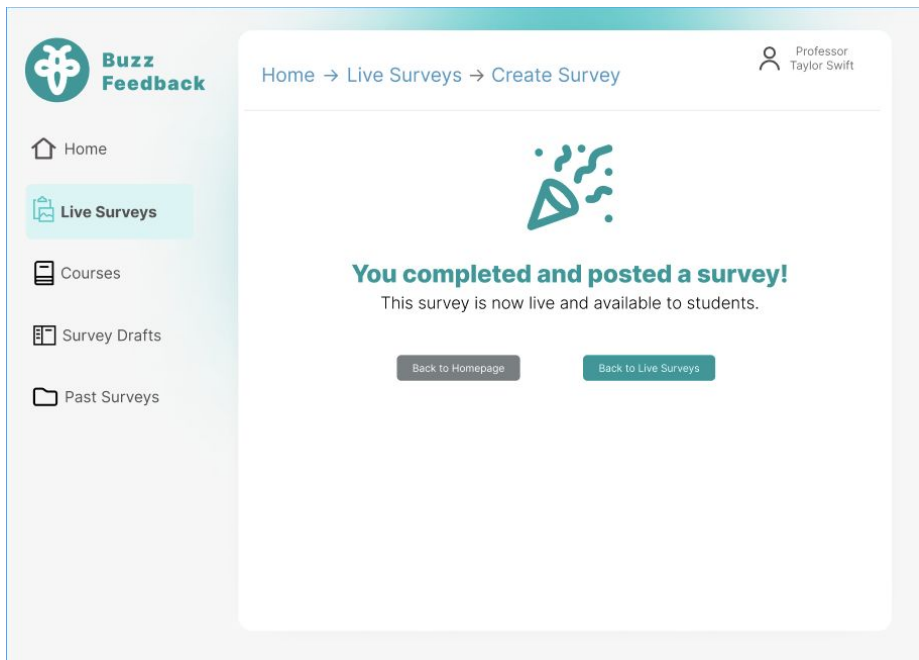
1 / 25
Submissions

Blank Survey

End of Semester Template

No Feedback after Actions

Solution: Added banners congratulating users for submitting/posting surveys



Confusing Meaning of Colors

Solution: Added legend to explain colors' meanings

Before - Question page

Q5 How would you describe the homework assignments?

Homework didn't match lecture materials Homework was fine Clear homework instructions

Confusing homework instructions Homework reinforced lecture materials

Tedious requirements for submissions Minimum Requirements

Difficult homework Easy homework

Q6 How would you describe the exams?

Confusing questions on exam Manageable Exam Matches with lecture materials

Difficult exam Clear questions on exam

Didn't match lecture material Easy exam

After - Question page

Q6 How would you describe the homework assignments?   

 Negative Comment  Positive Comment

Homework didn't match lecture materials  Clear homework instructions 

Confusing instructions  Homework reinforced lecture materials 

Tedious requirements for submissions  Minimum requirements 

Difficult homework  Homework was manageable 

Q7 How would you describe the exam?   

 Negative Comment  Positive Comment

Confusing questions on exam  Clear questions on exam 

Exam content didn't match lecture materials  Exam content reinforced lecture materials 

No Grouping of Elements on Page

Solution: Added grouping via proximity and boxes

Before - Question page

Q1 Overall, how would you rate this course?

- ☐ Excellent
- ☐ Very Good
- ☐ Good
- ☐ Fair
- ☐ Poor

Q2 Overall, how would you rate this instructor?

- ☐ Excellent
- ☐ Very Good
- ☐ Good
- ☐ Fair
- ☐ Poor

Q3 Overall, how useful were the resources (lecture recordings, textbook readings)?

- ☐ Extremely Useful
- ☐ Very Useful
- ☐ Somewhat useful
- ☐ Not so useful
- ☐ Not at all useful

Next

After - Question page

The screenshot shows the 'Buzz Feedback' interface for a survey titled 'Visualization End of Semester Survey'. The survey is for the '2023 Spring' semester and is due on '05/30/2023'. The user is identified as 'Student Liam Hemsworth'. The survey progress is shown on the left: 1. Course evaluation, 2. Instructor evaluation, 3. Materials evaluation, 4. Additional comments, and 5. Submit. The main content area shows two questions. Question 1 asks 'Overall, how would you rate the course?' with five radio button options: Excellent, Very Good, Good, Fair, and Poor. Question 2 asks 'Overall, how would you describe the course?' and features two columns of text input boxes. The left column is for 'Negative Comment' and the right column is for 'Positive Comment'. The negative comment boxes contain: 'Hard to follow', 'Boring', 'Not practical in real world', 'Poor explanation', and 'Low engagement'. The positive comment boxes contain: 'Slides are easy to follow', 'Interesting topics', 'Practical knowledge', 'Clear explanation', and 'Active classroom Environment'. At the bottom, there are buttons for 'Back', 'Save for later', and 'Next'.

No Method to Block Disrespectful Comments

Solution: Added a feature to block disrespectful comments

The screenshot displays the 'Buzz Feedback' application interface. On the left, a vertical sidebar contains five steps: 1 Course evaluation, 2 Instructor evaluation (highlighted), 3 Materials evaluation, 4 Additional comments, and 5 Submit. The main content area is titled 'Visualization End of Semester Survey' and includes the text '2023 Spring' and 'Due 05/30/2023'. It shows two survey questions: Q3 'Overall, how would you rate the instructor?' with radio button options (Excellent, Very Good, Good, Fair, Poor) and Q4 'Overall, how would you describe the instructor?' with a legend for Negative Comment (red) and Positive Comment (green). Below Q4, there are several tag buttons: 'Mean', 'Impatient', 'Messy', 'Low engagement', 'Shit', 'Kind', 'Organized', and 'Inconsistent'. A modal dialog titled 'Adding Tag Failed' is overlaid on the screen, displaying the warning: 'Warning: you cannot curse in the course evaluation.' and a 'Delete the tag' button. At the bottom of the interface, there are 'Back', 'Save for later', and 'Next' buttons.

Cannot Distinguish between Student's and Professor's View

Solution: Used green and blue to distinguish different views and write their names on the top

 **Buzz Feedback**

Dashboard

Professor Taylor Swift

Home

Courses

Live Surveys

Survey Drafts

Past Surveys

Welcome to the course evaluation site Buzz Feedback!

We provide a way to collect qualitative course evaluation feedback by utilizing a tagging system, keywords or phrases are used to categorize data.

This directs feedback to be more specific and constructive, leading to more helpful evaluations from students!

Courses


Human-Computer Interaction
2023 Spring


Visualization
2023 Spring


Mobile Networks
2023 Spring

 **Buzz Feedback**

Dashboard

Student Liam Hemsworth

Home

Courses

Live Surveys

Past Surveys

Welcome to the course evaluation site Buzz Feedback!

We provide a way to collect qualitative course evaluation feedback by utilizing a tagging system.

This helps feedback to be more specific and constructive, while also decreasing the time and effort to fill out these evaluations!

Note that your answers will be sent anonymously to professors except for demographic information.

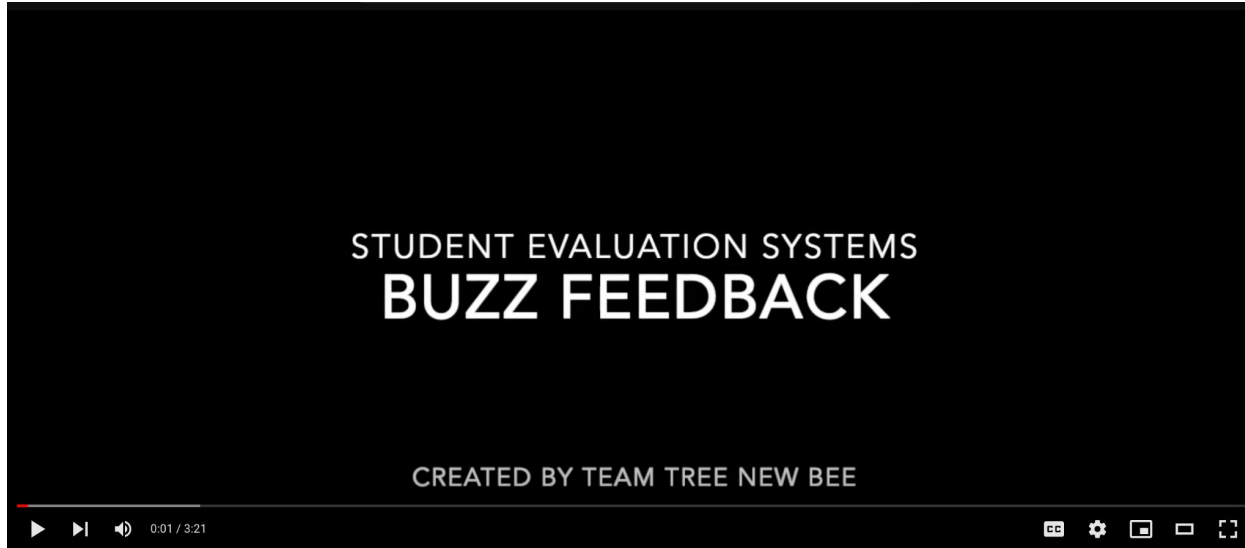
Courses


Human-Computer Interaction
2023 Spring


Visualization
2023 Spring

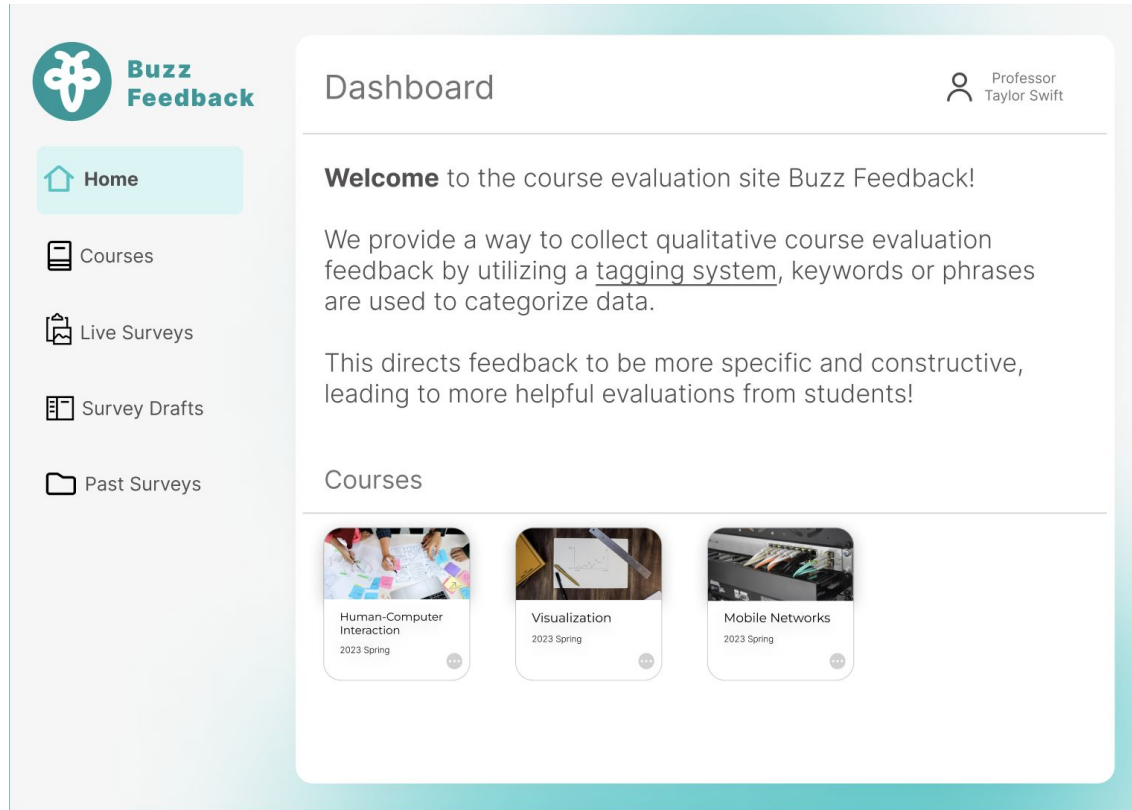

Mobile Networks
2023 Spring

Video Prototype



https://youtu.be/JiUZTGMIU_o

Figma Prototype



<https://www.figma.com/proto/iYWKIzLu9FEjgzUDt3JOIy/Hifi-redesign?type=design&node-id=191-4746&scaling=scale-down&page-id=0%3A1&starting-point-node-id=191%3A4746&show-proto-sidebar=1>

Portfolio

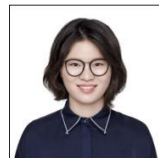
Tree New Bee

Welcome to **Tree New Bee**! We deliver a way to collect more honest, constructive student feedback on courses.

Meet the Team



Sammy Lee



Qingwen "Cecelia" Cheng



Joanne "Jo" Lee

1 Empathize

We interviewed users to find common needs and pain points.

POWERPOINT

2 Define

We used Empathy maps, POVs, and journey maps to analyze our findings from the user interviews.

POWERPOINT

3 Ideate

We generated ideas using methods such as "How Might We's and brainstorming to find the solution idea.

POWERPOINT

4 Low Fidelity Prototype

We used paper prototyping to build a lo-fi prototype and collected user feedback through think-aloud tests.

POWERPOINT

VIDEO

PROTOTYPE

5 High Fidelity Prototype

We created a hi-fi digital prototype and collected user feedback through think-aloud tests.

POWERPOINT

VIDEO

PROTOTYPE

6 Final Presentation

We applied from heuristic evaluations by our peers and presented all we learned throughout the semester.

POWERPOINT

VIDEO

PROTOTYPE

<https://www.figma.com/community/file/1234368361146463621>

Final Reflections and Takeaways

- It's easy to design a product with low discoverability because you forget what it was like not knowing about this long-term design project
- Walkthroughs are important for centering design on user experience rather than the functions/features of a product
- Sometimes, feedback points out a problem but not the best solution to it
- Designing for a computer screen rather than a mobile device increases the work by a magnitude

Appendix - Heuristic Evaluations Results P1

Heuristic	Reviewer	Violation	Recommendation	Severity	Problem code
1. Visibility of System Stats	Alice Frederick	After completing a student survey and hitting submit, it takes it back to the home screen without confirmation that it was completed.	On the student view, I think it would be useful to have a “past surveys” list like the instructor view so that the finished survey shows up on the dashboard as well (even though it disappears from the live surveys list).	2	No Feedback after Actions
	Ryan Chan	There is the ability to go immediately home on some pages and others you would continuously have to hit back. Additionally, on either type of question, I believe that it could be helpful if a student is able to leave a comment regarding anything else they might want to say that is not fully captured by the response or tags that they create.	Consider adding home buttons everywhere if the user wants to just quit out of everything instantaneously and go back to the main menu. Also, it could be useful to add a comment section at the end of any type of question.	3	Difficult Navigation, No Text Box Answer in Survey
	Ananya Barthakur	The app does not provide users with feedback or loading indicators when they perform certain actions.	Adding loading indicators after key actions like saving, submitting, adding a tag.	2	No Feedback after Actions
	Nikki Lockwood	The different pages of the application do not appear to have titles or markings indicating where a person is and what the purpose of the page is.	Add titles or clear markings to each page of the application so that the user is able to easily identify what task each page is meant to do.	3	Difficult Navigation
	Vivienne Kupiecki	After you submit a survey, there isn't an indication whether or not you added a new one.	After you complete a survey for a course, maybe have a temporary green banner that indicates that the survey has been added.	1	No Feedback after Actions
	Vivienne Kupiecki	Looking at the past surveys page, it is difficult to understand what the numbers indicate (e.g. what is average instructor score? Is it out of 5? 10?)	For the ratings, maybe have a denominator to indicate how ‘good’ or ‘bad’ the scores are.	1	Confusing Color, Confusing Survey Results Page
2. Match Between System and Real World	Joy Kondo	Some answer options (such as “not so useful”) seem vague and not very clear.	Might be a good idea to streamline the answer choices to be more conventional choices. Rather than having users enter the answer choices, they could be given a couple options to choose from when creating the survey.	3	No Templates

Appendix - Heuristic Evaluations Results P2

Heuristic	Reviewer	Violation	Recommendation	Severity	Problem code
2. Match Between System and Real World	Joy Kondo	There is no option to indicate the semester/year of the class that the professor is creating the survey for. There is also no option to set the deadline for the survey.	Create spaces for the semester/year and also the deadline for the surveys.	4	Missing Inputs for Create Survey
	Vivienne Kupiecki	Although the '+' icon appears to indicate to add a new survey, there's no clear indication for what you're supposed to do.	Maybe add a 'add new survey' title above the icon. The titles above the survey widgets make it clear what the app is supposed to do, but maybe add a short description of the application as well.	1	No Word Labels for Buttons
3. User Control and Freedom	Alice Frederick	In Professor (Task 2), in the drop down area to see in depth info about questions & student responses, it's not clear how to immediately get back to the summary page – home goes all the way home. Closing the top drop down simultaneously goes back but this isn't a conventional use of the drop down to also be a back button so the "exit" is hard to find.	Perhaps replace the home button with a back button so that the user goes back to the previous page they were on and can go to home from there. This is what I thought it would do. Other ideas could be breadcrumbing so someone can navigate to all screens quickly, too.	4	Difficult Navigation
	Alice Frederick	There is no exit from the add new survey page, if a professor does not want to make a survey they cannot exit/undo that action.	A back button with a message that progress would not be saved would be useful for exiting out of survey creation.	4	Difficult Navigation, No Error Checking
	Joy Kondo	Past surveys users are only able to see one question at a time and need to exit out to view a different question.	Make the page scrollable so that one action can be eliminated.	3	No Scrolling
	Joy Kondo	Users are unable to move the tags around.	The feedback might be more useful if users can rank their tags in order of importance or severity.	1	No Rating Tags
	Ananya Barthakur	There does not seem to be an exit button on the survey page.	Implement a feature that allows users to undo or reverse actions to provide more control and freedom to users.	3	Difficult Navigation
	Ryan Chan	<i>[shared with heuristic 1]</i> There is the ability...Additionally, on either type of question, I believe that it could be helpful if a student is able to leave a comment regarding anything else they might want to say that is not fully captured by the response/tags that they create.	Consider adding home buttons everywhere...Also, it could be useful to add a comment section at the end of any type of question.	3	Difficult Navigation, No Text Box Answer in Survey

Appendix - Heuristic Evaluations Results P3

Heuristic	Reviewer	Violation	Recommendation	Severity	Problem code
4. Consistency and Standards	Alice Frederick	The design uses orange = negative and blue = positive. I'd say the more consistent color choices to indicate this is red and green from what we see in many other applications that are similar. Even on this heuristic eval preset, the neg/pos is assumed to be red/green!	Change the color to more conventional colors that match the standard of negative/positive, or provide a header that indicates neg/pos feedback instead of assuming the user knows what the colors mean automatically. The legend is only visible on one screen during quiz creation.	1	Confusing Color
	Alice Frederick	Also, the "average rating" questions show up in orange when on other questions the color is supposed to indicate negative feedback.	Change the average ratings to not be one of your neg/pos connotation colors.	1	Confusing Color
	Joy Kondo	"Average" values on the past survey summary pages are not intuitive. What is the total score that the professor could have received?	It might be good to use a percentage or indicate that all the values are out of 10, etc.	2	Confusing Survey Results Page
	Joy Kondo	Standard questions/answer choices are unavailable.	Templates for questions (esp to phrase the questions well) and answer choices would be helpful.	1	No Templates
	Ryan Chan	In the midterm survey summary example it showed the average score of some questions and how many tags were clicked based on different questions, however it was missing a Q3.	I think for consistency purposes the summary should include all questions responses to some degree even if it is just mentioning something minor.	1	Confusing Survey Results Page
5. Error Prevention	Alice Frederick	Deletion of question occurs immediately, and is right next to the copy icon. It's easy to accidentally delete an entire question.	There's many solutions, some are: a warning message, an undo button, different placing of the copy/delete buttons on the screen to make the action more obviously intentional so slips occur less.	3	No Error Checking
	Alice Frederick	Confirmation of a submission to a survey is not present; there is no way for the student to review their answers before submitting.	A summary page of the student user's responses would be useful before they submit so they can double check their selections before handing it in to the professor.	3	No Error Checking
	Anaya Barthakur	The app does not have any error prevention mechanism to prevent users from accidentally deleting questions or tags.	Implement a confirmation dialog, to prevent users from accidentally deleting questions or tags.	2	No Error Checking

Appendix - Heuristic Evaluations Results P4

Heuristic	Reviewer	Violation	Recommendation	Severity	Problem code
5. Error Prevention	Joy Kondo	Deleting a question is only one click – could be easy to delete a question by mistake.	An “undo” button might be helpful, or a confirmation button before a question actually gets deleted.	4	No Error Checking
	Joy Kondo	Posting a survey is only one click so it could be prematurely posted.	Professors should have the ability to preview and check the survey before they post it or to edit it once it has been posted.	4	No Error Checking
	Vivienne Kupiecki	Having a ‘save’ confirmation is helpful for reducing accidental slips (e.g. the user accidentally adds the wrong tag).	Add a pop up window for deleting questions, such as “Are you sure you want to delete this question?” Also, for the tags, maybe have a text prompt for deleting tags as well.	2	No Error Checking
6. Recognition Rather Than Recall	Ananya Barthakur	Users may not remember which tags they used to categorize their questions or responses.	Perhaps using green or red colors for positive/negative feedback respectively, could help the instructor understand if the tag calls for improvement or not.	2	Confusing Colors
	Vivienne Kupiecki	The icons for filling the survey are easily recognized. such as the trash and edit icons. The train of use is a little difficult to understand.	Add a menu at the top of the application for additional functionality (they might not know that you can select the different surveys. Add another home page button so the user can go back during a survey completion.	2	Difficult Navigation
7. Flexibility and Efficiency of Use	Alice Frederick	For a more experienced user, adding each tag and typing them out individually could be tedious.	Perhaps add suggestions of tags to add when “Add tag” is selected, based on the user’s previous commonly made tags. Allows the professor to in a way “save” their commonly used tags to make the survey creation quicker across courses.	2	No Templates
	Joy Kondo	Users who create surveys do not have the option to save previous responses or re-create surveys that they created once.	Chances are professors would like to reuse surveys that they created in the past, with the same answer options (esp for the qualitative questions). Having a reusable template would be good.	1	No Templates
	Nikki Lockwood	The application doesn’t have any kind of short-cut or preformatted example for novice users.	Creating sample quizzes/questions that the user can defaultly select before editing/customizing their own.	2	No Templates

Appendix - Heuristic Evaluations Results P5

Heuristic	Reviewer	Violation	Recommendation	Severity	Problem code
8. Aesthetic & Minimalist Design	Alice Frederick	The summary page design could be more efficient, as right now it's a little bit difficult to glean all information quickly. I was a bit confused reading it at first & the improvements/good section is meshed between survey results.	Perhaps for each question, utilize bar graphs to indicate neg/pos feedback instead of splitting it and showing each question's result twice; it can be hard to compare them in the way they are right now since the heights of neg/pos can't be directly compared quickly. If you're going to use visual tools to help understand the data, I think a chart/graph would be more useful! Also, maybe put the improvements/goals apart from the survey results – I couldn't tell if these are suggestions made by the system you made or if they were student responses.	3	Confusing Survey Results Page
	Joy Kondo	"Improve" and "Good" are not the best titles because they aren't opposites. Not exactly sure these titles are relevant.	Use "Good and "Bad or "Positive and Negative to indicate what the feedback means.	4	Confusing Survey Results Page
	Nikki Lockwood	The different elements on each page are not obviously separated and aren't pleasing to the eye.	Place different content blocks onto colored rectangles or use a shadowing effect to show a clear difference.	1	No Grouping
	Ryan Chan	Good design that is very clean and effective, however I would consider changing the color of the tags.	As for the color of the tags, I personally would match more of a traffic light scheme (red (negative), yellow (neutral), green (positive)). This is obviously just a suggestion, but I think it would make it a lot easier to understand the difference between positive/negative reviews even if there was no legend. You could also retain a gray color for neutral if you so choose.	1	Confusing Color
	Vivienne Kupiecki	The number of students button at the top right of the survey is not relevant to a student (how would they know the number of students in their class?).	Delete the number of students button.	2	Unnecessary Feature
9. Help Users Recognize, Diagnose, and Recover from Errors	Alice Frederick	There are no error messages in the whole prototype; refer to my analysis in heuristic #5!	Refer to heuristic #5, which mentions recovery from error.	3	No Error Checking
	Joy Kondo	No user messages – view #5.	Add error messages	34	No Error Checking

Appendix - Heuristic Evaluations Results P6

Heuristic	Reviewer	Violation	Recommendation	Severity	Problem code
9. Help Users Recognize, Diagnose, and Recover from Errors	Ryan Chan	I am not sure if it is a possibility within your app itself, but it could be possible for a student to select multiple answers for multiple choice questions within your survey if they are not set to single response questions.	Ensure that this situation can not happen even if it may be obvious.	1	Confusing Survey Results Page
	Vivienne Kupiecki	There isn't a threshold (from what I understood) of the number of tags or questions required for each question. Therefore, students could fill out a survey with missing questions or less feedback than others.	If there needs to be a minimum amount of questions filled. Include error message. Have error prompts for questions that are added but aren't filled out.	2	Confusing Survey Results Page
10. Help and Documentation	Alice Frederick	From the instructor view, it's clear that the submissions are anonymous. I feel that on the student side, it doesn't indicate this in any way.	Some reassurance that the answers are kept anonymous to the professor on the student side would be useful documentation of that aspect so the user isn't biased in their response.	2	No Documentation
	Ananya Barthakur	The use of tags to categorize questions may not be intuitive for users, and they may not understand how to use this feature effectively.	Consider using more familiar and intuitive language to label the tags and provide clear instructions on how to use this feature effectively.	4	No Documentation
	Joy Kondo	Summary page for past surveys: unclear why some things are orange, some are blue. Why are some question numbers repeated?	Some documentation or explanation might help here so that users will know exactly what all the color encodings and text refer to.	3	No Documentation
	Joy Kondo	It definitely would have been helpful to see a walkthrough video going through the student view as well. I am having a hard time determining all the features because I'm not completely sure what the experience is like or the student.	Include student view in walkthrough.	3	No Documentation

Appendix - Heuristic Evaluations Results P7

Heuristic	Reviewer	Violation	Recommendation	Severity	Problem code
10. Help and Documentation	Nikki Lockwood	There isn't any help page or instructions to help the user figure out what to do.	Provide a help tab and include instruction buttons on the create quiz page.	3	No Documentation
	Ryan Chan	Even though it may appear obvious to the standard student/professor how to utilize the app, there should be more directions in making sure the students can follow the directions of the respective survey which in turn limits the potentiality for mistakes. This includes answering questions and creating questions as well.	Provide basic but informative instructions for both the student/professor view so they can seamlessly go about utilizing the features in the app in order to create/answer surveys without any hassle.	2	No Documentation
	Vivienne Kupiecki	There is very little documentation on the application, but because of the straightforwardness of the application, it may not be necessary.	Be more explicit of what students and instructors have access to. Can students look at the results of past surveys? Should that be an option for them? Or should they just be able to see their past answers?	1	No Documentation
	Alice Frederick	From the instructor view, it's clear that the submissions are anonymous. I feel that on the student side, it doesn't indicate this in any way.	Some reassurance that the answers are kept anonymous to the professor on the student side would be useful documentation of that aspect so the user isn't biased in their response.	2	No Documentation

Appendix - Heuristic Evaluations Report

Ave. Severity	Problem Code	Recommendations	Heuristics
4	Missing inputs for creating survey	• Make inputs for semester, year, deadline	2
3.2	Difficult navigation	• Home buttons and exit buttons everywhere • Include top menu • Include name of current page at top of screen	1,3,6
3	No text box answer in survey	• Include a text box answer for any last comments	1
3	No scrolling	• Add scrolling feature	3
2.9	No error checking	• ‘Are you sure’ messages before deleting questions • ^ for posting/submitting surveys	5,9
2.5	No documentation	• Include documentation for both users	10
2.5	Confusing survey results page	• Change “Good”/”Improve” section to “Positive”/ ”Negative” • Use bar graphs to show multiple choice answers, pie charts for tags • Show all the questions’ results on one page	4,8
2	Unnecessary feature	• Take out the asking for # of students question on student side	8
1.8	No templates	• Offer template for survey creation	3
1.7	No feedback after action	• Show loading indicators after deleting/creating stuff • Show green banner after professor creates survey • Show students’ submitted surveys in ‘Past Surveys’ section	1
1.6	Confusing color meanings	• Change tag colors to green (positive), red (negative) • Show quantitative data in colors different from tags	1,4,6,8,10
1	No word labels for buttons	• Add ‘Create Survey’ to the ‘+’ button	2
1	No rating tags	• Allow students to move around tags to rate relevancy	3
1	Limits to multiple choice question	• Set limits to how many answers can be selected in multiple choice question	9
1	No grouping	• Put elements of page in blocks or add shadowing	8

Appendix - Implemented Revisions

Problem Code	Revision or Reason to Disregard
Missing inputs for creating survey	Added deadline input when creating survey, semester/class gets filled in via a backend database for professor accounts
Difficult navigation	Added breadcrumbing, a general menu on the side, and home button
No text box answer in survey	Added text box questions
No scrolling	Added scrolling
No error checking	Added error messages for deleting questions, a 'Submit' page that allows professors and students to check their survey before posting/submitting
No documentation	Added homepage that explain tagging and privacy policy
Confusing survey results page	Put survey results all on one page, used bar graphs to show data
Unnecessary feature	Removed question asking for # of students in class for the student view
No templates	'Create Survey' button offers options of blank document or default template
No feedback after action	Added 'Thank you for submitting/posting' page for when a professor submits a survey or when a student deletes a survey
Confusing color meanings	Set colors to green (positive), red (negative), grey (neutral)
No word labels for buttons	Added word labels for most buttons except the home button
No rating tags	<i>Did not implement because the ratings of tags for an individual, anonymous student didn't seem useful for professors, and the frequency of tags among the whole class already rates the tags by relevance</i>
Limits to multiple choice question	<i>Did not implement because mult. choice questions already takes 1 answer</i>
No grouping	Grouped questions in boxes in both live surveys and results page